AITM LAB 2: ARP Cache Spoofing Attack

Man-in-the-Middle Interception via Docker Containers

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EXECUTIVE SUMMARY

Executive Summary

Objective

Demonstrate ARP poisoning to position as AitM and intercept plaintext traffic

Environment

Docker-based SEED Labs (Ubuntu 20.04, Containers A: 10.9.0.5, B: 10.9.0.6)

Key Findings

Successfully poisoned ARP cache; intercepted plaintext netcat/telnet traffic

Security Implication

Plaintext protocols (telnet, nc) are vulnerable to network-level eavesdropping

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LAB SETUP

Lab Setup & Architecture

Network Topology diagram: (show 10.9.0.0/24 subnet with hosts A, B, gateway 10.9.0.1)

Docker containers: launched via docker-compose.yml

Tools used: arpspoof (dsniff), tcpdump, netcat, arp/ip neigh commands

Attack flow: A (attacker) → poison B's ARP → intercept traffic

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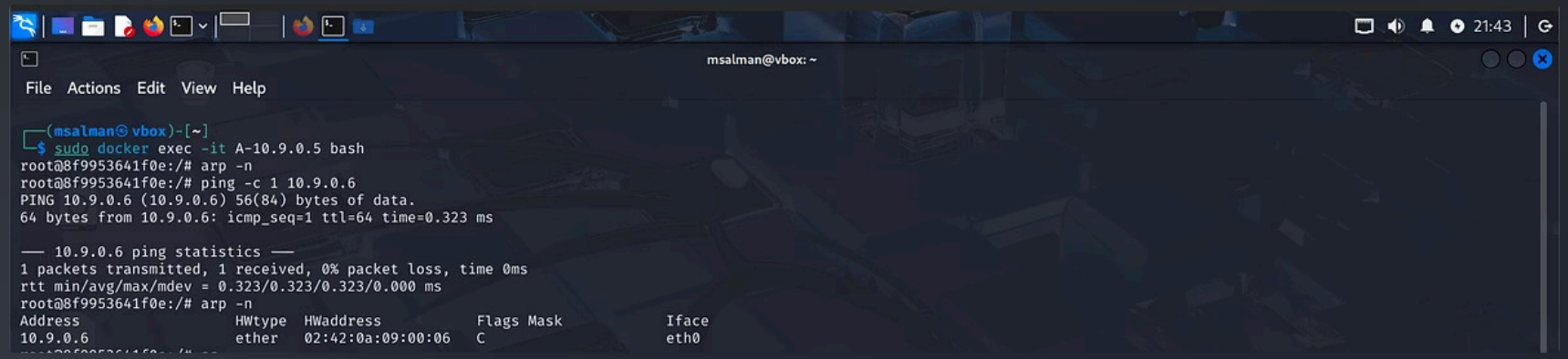

TASK 1

Task 1 - Baseline ARP (Before Attack)

Description: Host A captures ARP table before spoofing

Key Entries:

- 10.9.0.6 (Host B): MAC 08:00:27:a4:3e:9c (correct)
- 10.9.0.1 (Gateway): MAC 08:00:27:75:8b:bf (correct)

Ping successful: 1 packet transmitted, 0% loss

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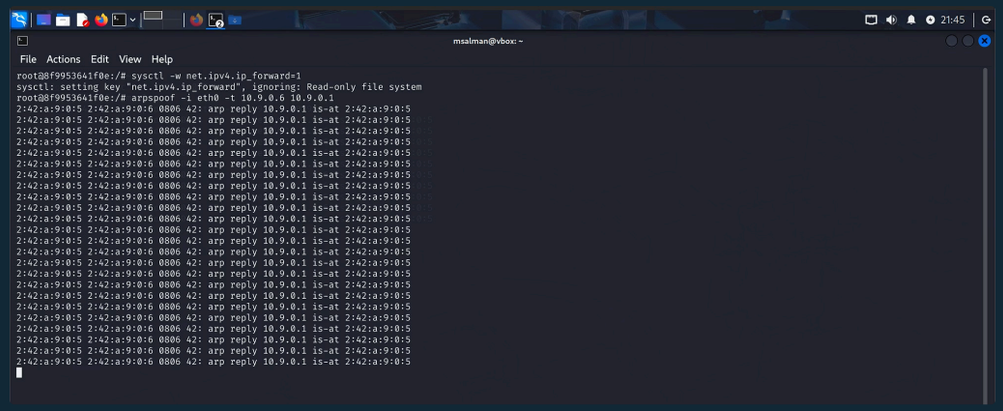

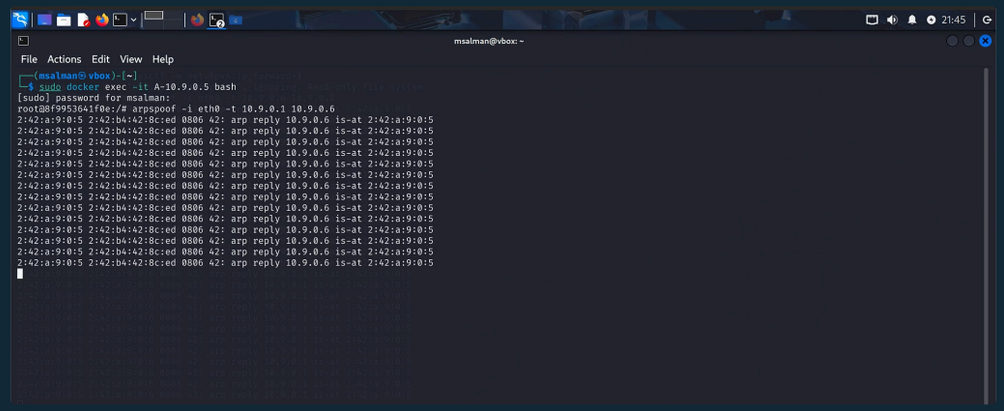

TASK 2

Task 2 - ARP Spoofing Execution

Description: Attacker A runs symmetric ARP spoofing with IP forwarding enabled

Commands:

- sysctl -w net.ipv4.ip_forward=1
- Terminal 1: arpspoof -i eth0 -t 10.9.0.6 10.9.0.1
- Terminal 2: arpspoof -i eth0 -t 10.9.0.1 10.9.0.6

Output shows: continuous gratuitous ARP replies (every ~1-2 seconds)

Attacker MAC spoofed: 2:42:a:9:0:5 (Docker container A)

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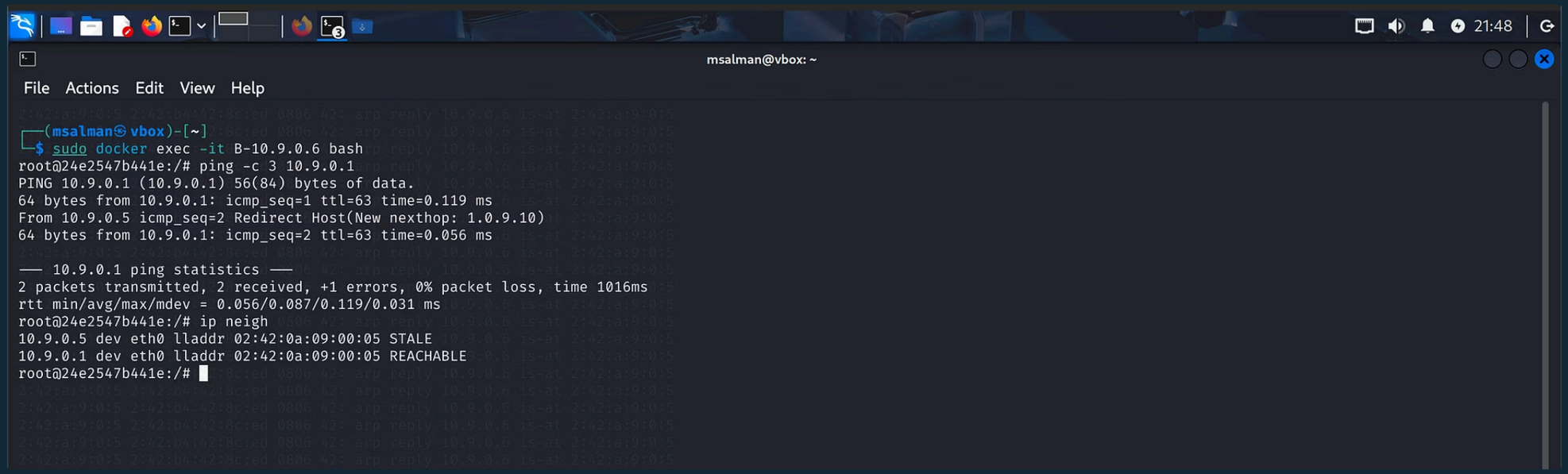

TASK 3

Task 3 - Verification (Poisoned ARP Cache on B)

Description: Host B's ARP cache after poisoning attack

Critical Finding: Both 10.9.0.1 (gateway) and 10.9.0.5 (attacker) now map to A's MAC:

- 10.9.0.5 (Attacker A): 02:42:0a:09:00:05 ✓ Correct
- 10.9.0.1 (Gateway): 02:42:0a:09:00:05 ✓ SPOOFED—originally 08:00:27:75:8b:bf

All B↔️Gateway traffic now routes through A (AitM achieved)

Flags: "REACHABLE" confirms cache populated

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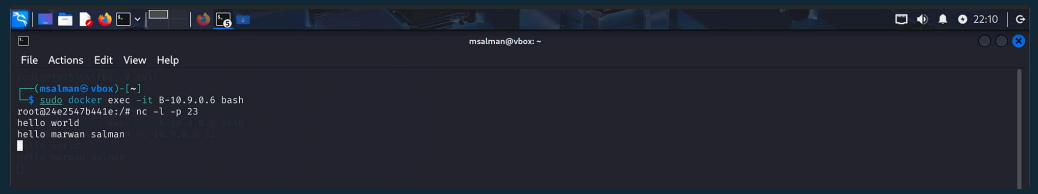

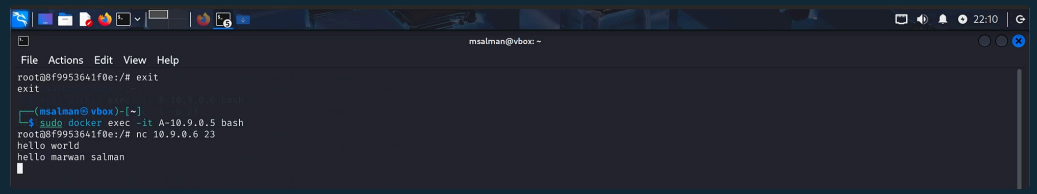

TASK 4

Task 4 - Netcat Traffic Interception (Demo)

Description: Client A sends plaintext data to Server B via poisoned ARP path

Netcat Demo (Alternative to Telnet):

- Server B: nc -l -p 23 (listening on telnet port)
- Client A: nc 10.9.0.6 23 (connects through AitM)
- Plaintext transmitted: "hello world" + "hello marwan salman"
- All data visible in cleartext (no encryption)

Why netcat instead of telnet: Telnet unavailable; netcat provides identical plaintext behavior

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ANALYSIS

Security Analysis & Implications

1

Vulnerability Chain

- ARP lacks authentication (gratuitous ARP accepted without verification)
- Attacker poisons target ARP cache via forged replies
- All traffic destined for gateway routes through attacker
- Plaintext protocols (telnet, HTTP, FTP) fully exposed

2

Risk Severity: HIGH

- Confidentiality: BROKEN (all traffic readable)
- Integrity: BROKEN (attacker can modify packets)
- Availability: BROKEN (attacker can drop/DoS traffic)

3

Real-world impact

Credential theft, MITM web requests, session hijacking

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MITIGATION

Mitigation & Defense Strategies

1

Static ARP entries

Manual ARP table population (inconvenient at scale)

2

ARP spoofing detection

Monitor for duplicate MACs or rapid ARP changes

3

Protocol-level

Use SSH (encrypted) instead of telnet

4

Network-level

- VLAN isolation + DHCP snooping
- Dynamic ARP Inspection (DAI) on switches
- ARP monitoring tools (ArpWatch, Suricata)

Lab lesson: Always encrypt sensitive data (SSH, HTTPS, TLS)

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CONCLUSION

Conclusion & Key Takeaways

- Successfully demonstrated ARP cache poisoning → AitM attack
- Intercepted plaintext traffic (netcat) under controlled lab environment
- Plaintext protocols fundamentally insecure at L2 without encryption
- Encryption (TLS/SSH) mandatory for confidential data
- SEED Labs provides hands-on understanding of network layer vulnerabilities
- Future work: Extend to DNS spoofing (Bonus AITM 3), test with HTTP interception

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